Project: Worked examples
Title: Sealed heating system
Status: DRAFT · Author: TempestOS (2026-10-23) · Checker: — (—) · Issued: —

1. Scope and method

Calculates the catalogue output each room's emitter needs at the design flow, return and room temperatures, and checks it against the selected emitters. It also gives the minimum expansion vessel for the system volume, temperatures, static head and pressures.

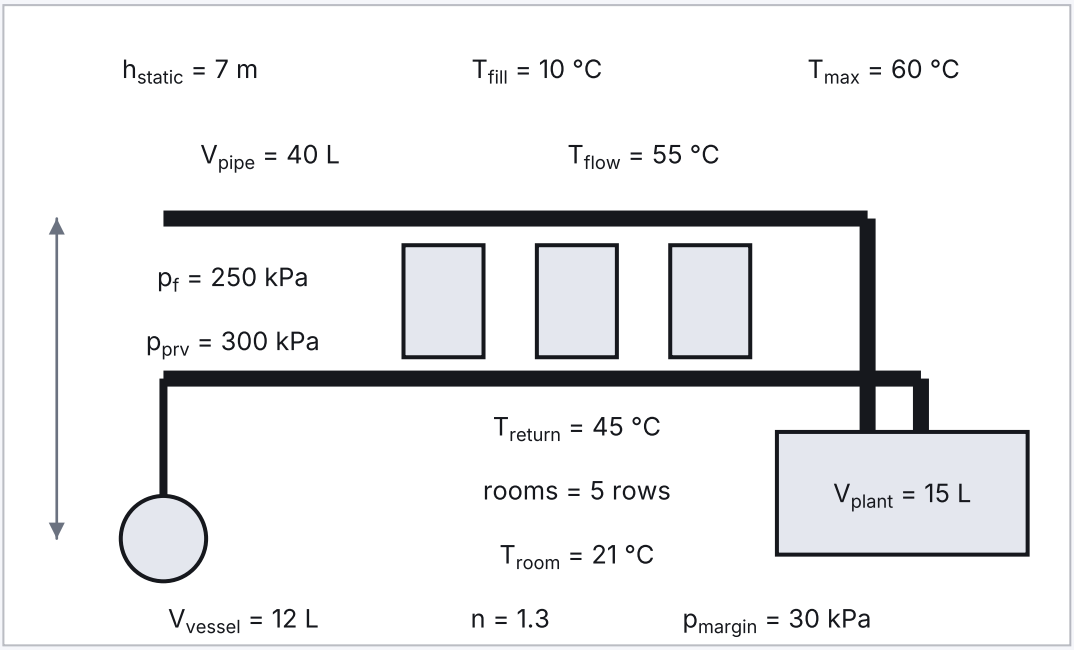


Figure 1 — A sealed heating circuit: the boiler, the flow and return pipework, the emitters of the rooms as listed (three drawn), and the expansion vessel below the static height. Not to scale; the labels are the inputs as entered.

WHAT IT CALCULATES	RELATION
Emitter temperature difference	$\Delta T = (T_{flow} + T_{return})/2 - T_{room}$, or the log mean $(T_{flow} - T_{return})/\ln((T_{flow} - T_{room})/(T_{return} - T_{room}))$
Output at design	$\Phi = \Phi 50 \cdot (\Delta T/50)^n$; required catalogue output $\Phi 50 = \Phi_{req}/(\Delta T/50)^n$
System volume	$V_s = \Sigma V_{em} + V_{pipe} + V_{plant}$
Water expansion	$e_x = \rho(T_{fill})/\rho(T_{max}) - 1$ (water density by Kell 1975), or a glycol record
Vessel pre-charge	$p_i = \rho \cdot g \cdot h_{static} + p_{margin}$ (at least the static head)
Vessel size	$V_e = e_x \cdot V_s/[1 - (p_i + 1)/(p_f + 1)]$ (pressures in bar gauge, + 1 bar for absolute); the expansion volume only, sized for the higher of T_{max} and T_{flow}; a water reserve and a safety-valve allowance are not included

Basis: Radiators and convectors — test methods and rating (characteristic equation)[1](cl. 5); Heating — sealed systems and expansion[2](§1.4, 2016 ed.).

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
dT_{method}	Mean temperature difference	arithmetic mean (catalogue rating)		assumption: Catalogue ratings use the arithmetic mean.	any
rooms	Rooms and emitters (optional; blank = no emitter check)	5 row(s)		spec: Room heat loss calculation (B4) and radiator survey	any
h_{static}	Static height above the vessel	7	m	drawing: M-SEC-001 rev A	0 … 40 m
V_{pipe}	Pipework water content	40	L	assumption: 15 mm and 22 mm copper, about 120 m.	0 … 5000 L
V_{plant}	Plant water content (boiler, cylinder coil, buffer)	15	L	spec: Heat pump and plate exchanger content, datasheet	0 … 5000 L
V_{vessel}	Selected vessel volume (optional; blank = no vessel check)	12	L	spec: Site survey: existing 12 L vessel	1 … 5000 L
T_{flow}	Flow temperature	55	°C	spec: Heat pump design flow temperature	40 … 85 °C
T_{return}	Return temperature	45	°C	spec: Heat pump design return temperature	35 … 80 °C
T_{room}	Room temperature	21	°C	spec: Living-room design temperature	16 … 23 °C
T_{fill}	Fill temperature	10	°C	assumption: Mains water at fill.	4 … 30 °C
T_{max}	Maximum system temperature	60	°C	spec: Heat pump maximum flow temperature	30 … 95 °C
p_f	Final (maximum working) pressure, gauge	250	kPa	assumption: 50 kPa below the safety valve.	50 … 600 kPa
p_{prv}	Safety valve setting, gauge	300	kPa	spec: Heat pump safety valve	200 … 550 kPa
n	Emitter exponent	1.3		assumption: Panel radiators, manufacturer exponent about 1.3.	1.2 … 1.4

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
p_{margin}	Pre-charge allowance over static head	30	kPa	assumption: Usual allowance over static head.	0 … 100 kPa

TABLE ROW	VALUES
rooms[1]	name = Lounge; Phi = 1800 W; Phi50 = 3600 W; V_em = 12 L
rooms[2]	name = Kitchen; Phi = 1100 W; Phi50 = 2200 W; V_em = 8 L
rooms[3]	name = Bed 1; Phi = 900 W; Phi50 = 1800 W; V_em = 7 L
rooms[4]	name = Bed 2; Phi = 700 W; Phi50 = 1400 W; V_em = 5 L
rooms[5]	name = Bathroom; Phi = 600 W; Phi50 = 1100 W; V_em = 4 L

3. Assumptions

A1. The system is a closed LTHW system below 100 °C. (Catalogue ratings at 75/65/20 °C.)
A2. Emitter output follows Φ50·(ΔT/50)^n with the manufacturer's n. (Catalogue ratings at 75/65/20 °C.)
A3. Water density comes from the Kell (1975) correlation, within about 0.01 % of IAPWS-95 at atmospheric pressure. (Formula, not a table.)
A4. Glycol mixtures take the expansion coefficient e from a record. (Formula, not a table.)
A5. Gauge pressures are made absolute by adding 1 bar, as the vessel relation is usually written. (Within 1.3 % of the standard atmosphere.)
A6. V_e is the expansion volume by the CIBSE heating guide's relation only. (Stated omission.)
A7. V_e does not include a water reserve for system losses. (Stated omission; add it when the vessel is selected, from the heating-system standard in use.)
A8. V_e does not include a margin between the final pressure and the safety valve setting beyond p_f below p_{prv}. (Stated omission; add it when the vessel is selected, from the heating-system standard in use.)
A9. dT_{method} (Mean temperature difference) is taken as arithmetic mean (catalogue rating). (Catalogue ratings use the arithmetic mean.)
A10. V_{pipe} (Pipework water content) is taken as 40 L. (15 mm and 22 mm copper, about 120 m.)
A11. T_{fill} (Fill temperature) is taken as 10 °C. (Mains water at fill.)
A12. p_f (Final (maximum working) pressure, gauge) is taken as 250 kPa. (50 kPa below the safety valve.)
A13. n (Emitter exponent) is taken as 1.3. (Panel radiators, manufacturer exponent about 1.3.)
A14. p_{margin} (Pre-charge allowance over static head) is taken as 30 kPa. (Usual allowance over static head.)

4. Working

[14] $dT = (T_{flow} + T_{return}) / 2 - T_{room}$ [1](cl. 5)
$= (55\ degC + 45\ degC) / 2 - 21\ degC$
$= 29\ K$
Mean water-to-room temperature difference

[15] $dT_{fr} = T_{flow} - T_{return}$
$= 55\ degC - 45\ degC$
$= 10\ K$
Flow-to-return temperature difference (what the water gives up across the emitters)

[16] $T_m = (T_{flow} + T_{return}) / 2$
$= (55\ degC + 45\ degC) / 2$
$= 50\ degC$
Mean water temperature, (flow + return)/2

[17] f_{dT} = (dT / 50 K)^n
= (29 K / 50 K)^1.3
= 0.49256
Output correction from ΔT 50 K
[18] R1 = Φ_{req}/f_{dT}
= 1800 / 0.49256
= 3.6544 kW
Lounge: catalogue output needed; selected 3600 W gives 1773.2 W (utilisation 1.02)
[19] R2 = Φ_{req}/f_{dT}
= 1100 / 0.49256
= 2.2332 kW
Kitchen: catalogue output needed; selected 2200 W gives 1083.6 W (utilisation 1.02)
[20] R3 = Φ_{req}/f_{dT}
= 900 / 0.49256
= 1.8272 kW
Bed 1: catalogue output needed; selected 1800 W gives 886.6 W (utilisation 1.02)
[21] R4 = Φ_{req}/f_{dT}
= 700 / 0.49256
= 1.4212 kW
Bed 2: catalogue output needed; selected 1400 W gives 689.58 W (utilisation 1.02)
[22] R5 = Φ_{req}/f_{dT}
= 600 / 0.49256
= 1.2181 kW
Bathroom: catalogue output needed; selected 1100 W gives 541.81 W (utilisation 1.11)
[23] Φ_{tot} = Σ Φ_{req}
= 5100 W
= 5.1 kW
Total heat required by the rooms
[24] V_{em} = Σ V_{em}
= 36 L
= 36 L
Emitter water content
[25] V_s = V_{em} + V_{pipe} + V_{plant}
= 36 L + 40 L + 15 L
= 91 L
System water volume
[26] ρ_{fill} = ρ(T_{fill}), Kell 1975
= T = 10 °C
= 999.7 kg/m³
Water density at fill
[27] ρ_{max} = ρ(T_{max}), Kell 1975
= T = 60 °C
= 983.2 kg/m³
Water density at maximum temperature
[28] e_x = ρ_{fill} / ρ_{max} - 1
= 999.7 kg/m³ / 983.2 kg/m³ − 1
= 0.016783
Expansion coefficient of water
[29] p_{st} = ρ_{fill} * 9.80665 m/s^2 * h_{static}
= 999.7 kg/m³ × 9.80665 m/s^2 × 7 m
= 68.626 kPa
Static head at the vessel

[30] $p_i = p_{st} + p_{margin}$
= 68.626 kPa + 30 kPa
= 98.626 kPa
Pre-charge (initial) pressure, gauge
[31] $V_e = e_x * V_s / (1 - (p_i + 1\ bar) / (p_f + 1\ bar))$ [2](§1.4)
= 0.016783 × 91 L / (1 - (98.626 kPa + 1 bar) / (250 kPa + 1 bar))
= 3.5312 L
Minimum expansion vessel volume
[32] $\Phi_{req,gov} = \Phi_{req}$ of the governing room
= the room with the highest $\Phi_{req}/(\Phi 50 \cdot f_{dT})$
= 0.6 kW
Heat required by the governing room
[33] $\Phi_{out,gov} = \Phi 50 \cdot f_{dT}$ of the governing room
= its selected emitter at the design temperatures
= 0.54181 kW
Output of the selected emitter of the governing room at the design temperatures
[34] u_{emit} = max over rooms $\Phi_{req}/(\Phi 50 \cdot f_{dT})$
= worst room
= 1.1074
Emitter utilisation (governing room)
[36] $u_{vessel} = V_e / V_{vessel}$
= 3.5312 L / 12 L
= 0.29427
Vessel utilisation

5. Results

SYMBOL	VALUE	DESCRIPTION
Φ_{tot}	5.100 kW	Total heat required
V_s	91 L	System water volume
V_e	3.531 L	Minimum vessel volume
dT	29 K	Mean temperature difference
dT_{fr}	10 K	Flow-to-return temperature difference
T_m	50 °C	Mean water temperature
f_{dT}	0.49256	Output correction factor
R1	3.654 kW	Lounge: Φ50 needed
R2	2.233 kW	Kitchen: Φ50 needed
R3	1.827 kW	Bed 1: Φ50 needed
R4	1.421 kW	Bed 2: Φ50 needed
R5	1.218 kW	Bathroom: Φ50 needed
V_{em}	36 L	Emitter water content
e_x	0.016783	Expansion coefficient
p_i	98.63 kPa	Pre-charge pressure
u_{emit}	1.11	Emitter utilisation (governing)
u_{vessel}	0.29	Vessel utilisation

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
FAIL	Selected emitters against room heat (governing room)	0.6000 kW	0.5418 kW	1.11	
PASS	Selected vessel against minimum volume	3.531 L	12 L	0.29	[2](§1.4)

7. Applicability

WARNING n: a tabulated value not taken from a reference record (Emitter rating (exponent)); enter it as one so its source, edition and licence travel with it. [record.n]

8. Verification

Template bs.sealed-system@1.5.1 Content hash 0b317f14c51ed603 Engine calc-engine/1.0.0
Known-answer cases passed 4/4 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0

9. References

1. BS EN 442-2:2014.
2. CIBSE Guide B1.